



Lab 04

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Subject	AI

Classes In Python

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```
class Car:
    def __init__(self, brand, model, year):
        self.brand=brand
        self.model=model
        self.year=year

    my_car = Car("Toyota", "Camry", 2022)

    print(f"Brand: {my_car.brand}")
    print(f"Model: {my_car.model}")
    print(f"Year: {my_car.year}")
```



```
Brand: Toyota
Model: Camry
Year: 2022
```



```
class Car:
    def __init__(self, brand, model, year):
        self.brand = brand
        self.model = model
        self.year = year

    def display_info(self):
        print(f"Car Details: {self.year} {self.brand} {self.model}")

my_car = Car("Toyota", "Camry", 2022)

my_car.display_info()
```



```
Car Details: 2022 Toyota Camry
```

```

class Car:
    def __init__(self, brand, model, year):
        self.brand = brand
        self.model = model
        self._year = year

    @property
    def year(self):
        return self._year

    @year.setter
    def year(self, new_year):
        if 1886 <= new_year <= 2024:
            self._year = new_year
        else:
            raise ValueError("Invalid year! Year must be between 1886 and 2024.")

    def display_info(self):
        print(f"Car Details: {self._year} {self.brand} {self.model}")

my_car = Car("Toyota", "Camry", 2022)

print(f"Current year: {my_car.year}")

my_car.year = 2023
print(f"Updated year: {my_car.year}")

try:
    my_car.year = 1800
except ValueError as e:
    print(f"Error: {e}")

my_car.display_info()

```

Current year: 2022

Updated year: 2023

Error: Invalid year! Year must be between 1886 and 2024.

Car Details: 2023 Toyota Camry



```
class Car:
    def __init__(self, brand, model, year):
        self.brand = brand
        self.model = model
        self._year = year

    @property
    def year(self):
        return self._year

    @year.setter
    def year(self, new_year):
        if 1886 <= new_year <= 2024:
            self._year = new_year
        else:
            raise ValueError("Invalid year! Year must be between 1886 and 2024.")

    def display_info(self):
        print(f"Car Details: {self._year} {self.brand} {self.model}")

class ElectricCar(Car):
    def __init__(self, brand, model, year, battery_capacity):
```



```
        super().__init__(brand, model, year)
        self.battery_capacity = battery_capacity

    def battery_info(self):
        print(f"Battery Capacity: {self.battery_capacity} kWh")

    def display_info(self):
        super().display_info()
        self.battery_info()

my_tesla = ElectricCar("Tesla", "Model S", 2023, 100)

print(f"Brand: {my_tesla.brand}")
print(f"Year: {my_tesla.year}")

my_tesla.battery_info()

my_tesla.display_info()
```



```
Brand: Tesla
Year: 2023
Battery Capacity: 100 kWh
Car Details: 2023 Tesla Model S
Battery Capacity: 100 kWh
```

```

class Car:
    def __init__(self, brand, model, year):
        self.brand = brand
        self.model = model
        self._year = year

    @property
    def year(self):
        return self._year

    @year.setter
    def year(self, new_year):
        if 1886 <= new_year <= 2024:
            self._year = new_year
        else:
            raise ValueError("Invalid year! Year must be between 1886 and 2024.")

    def __str__(self):
        return f"{self._year} {self.brand} {self.model} (Car)"

    def display_info(self):
        print(self)

```

```

class ElectricCar(Car):
    def __init__(self, brand, model, year, battery_capacity):
        super().__init__(brand, model, year)
        self.battery_capacity = battery_capacity

    def __str__(self):
        return (f"{super().__str__().replace('(Car)', '')} "
                f"(Electric Car, {self.battery_capacity}kWh)")

    def battery_info(self):
        print(f"Battery Capacity: {self.battery_capacity}kWh")

regular_car = Car("Toyota", "Camry", 2022)
electric_car = ElectricCar("Tesla", "Model 3", 2023, 75)

print(regular_car)
print(electric_car)

regular_car.display_info()
electric_car.display_info()

```

```

→ 2022 Toyota Camry (Car)
   2023 Tesla Model 3 (Electric Car, 75kWh)
   2022 Toyota Camry (Car)

```

Function in Python

```
def calculate_sum(a, b):  
    return a + b  
  
sum = calculate_sum(5, 10)  
print(sum)
```

15

```
def greet(name="Guest"):  
    """Greets the user by name or as Guest if no name is provided"""  
    print(f"Hello, {name}!")  
|  
greet("Alice")  
greet()  
greet("Bob")  
greet("")
```

```
⇒ Hello, Alice!  
Hello, Guest!  
Hello, Bob!  
Hello, !
```



```
numbers = [2, 4, 6, 8]
squared_numbers = list(map(lambda x: x ** 2, numbers))
print(squared_numbers) # Output: [4, 16, 36, 64]
```



```
[4, 16, 36, 64]
```



```
def factorial(n):
    if n == 0 or n == 1:
        return 1
    else:
        return n * factorial(n - 1)
print(factorial(5))
```



```
120
```

Loops in Python



```
student_scores = {'Alice': 85, 'Bob': 78, 'Charlie': 92}

for student, score in student_scores.items():
    print(f"{student}: {score}")
```



```
Alice: 85
Bob: 78
Charlie: 92
```

